

main.c



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Run

```
1 #include <stdio.h>
2 #include <limits.h>
3 int main() {
4     int n, i, time = 0, completed = 0;
5     printf("Enter number of processes: ");
6     scanf("%d", &n);
7     int bt[n], rt[n], pr[n], wt[n], tat[n];
8
9     printf("Enter Burst Time and Priority:\n");
10    for(i = 0; i < n; i++) {
11        scanf("%d %d", &bt[i], &pr[i]);
12        rt[i] = bt[i];
13    }
14    while(completed != n) {
15        int min = INT_MAX, idx = -1;
16
17        for(i = 0; i < n; i++) {
18            if(rt[i] > 0 && pr[i] < min) {
19                min = pr[i];
20                idx = i;
21            }
22        }
23        rt[idx]--;
24        time++;
25
26        if(rt[idx] == 0) {
27            completed++;
28            tat[idx] = time;
29            wt[idx] = tat[idx] - bt[idx];
30        }
31    }
32    printf("\nProcess\tBT\tWT\tTAT\n");
33    for(i = 0; i < n; i++) {
34        printf("%d\t%d\t%d\t%d\n", i+1, bt[i], wt[i], tat[i]);
35    }
36    return 0;
37 }
```

## Output

Enter number of processes: 4

Enter Burst Time and Priority:

4 5

6 4

1 8

2 1

| Process | BT | WT | TAT |
|---------|----|----|-----|
|---------|----|----|-----|

|   |   |   |    |
|---|---|---|----|
| 1 | 4 | 8 | 12 |
|---|---|---|----|

|   |   |   |   |
|---|---|---|---|
| 2 | 6 | 2 | 8 |
|---|---|---|---|

|   |   |    |    |
|---|---|----|----|
| 3 | 1 | 12 | 13 |
|---|---|----|----|

|   |   |   |   |
|---|---|---|---|
| 4 | 2 | 0 | 2 |
|---|---|---|---|

=== Code Execution Successful ===